A Subquadratic Algorithm for Two-List Stable Matching

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Abstract

In the two-list stable matching problem, each of n men ranks the women by one of two master lists, and each of n women ranks the men by one of two master lists. The input has size $O(n)$. We give a deterministic algorithm that runs in $O(n^2\sqrt{\log\log n/\log n})$ time on the word RAM, providing a worst-case $o(n^2)$ bound for the problem.

The four groups of agents form a cycle of mutually acceptable groups. We unroll this cycle into a path of $4n+2$ copies of the instance, called sheets. The resulting market has a unique stable matching, obtained by repeatedly matching mutual first choices. Our structural theorem shows that the men of its middle sheet project to a stable matching of the original instance. The proof finds a stationary slice in a longer cover, bounds the two ends by a flow and winding argument, and splices out the excess middle. A Four-Russians implementation then batches the mutual-first-choice eliminations. The matching returned is the unique stable matching with the fewest pairs between the type-0 men and the type-1 women.

1 Introduction

In the stable marriage problem there are n men and n women, each with a strict ranking of all members of the other side. A perfect matching is *stable* if no man and woman prefer each other to their partners. Gale and Shapley (1962) showed that a stable matching always exists and can be found in $O(n^2)$ time. When preferences are given explicitly, this is optimal: Ng and Hirschberg (1990) proved that $\Omega(n^2)$ queries to the preference lists are needed, and Gonczarowski et al. (2019) proved a matching communication lower bound.

In many applications, however, preferences are *succinct*: agents rank the other side by a few shared criteria, so the input is much smaller than n^2. Moeller et al. (2016), and the journal version by Künnemann et al. (2019), asked whether such representations make stable matching easier. In their *d-list model*, every man ranks the women by one of at most d master lists, and every woman ranks the men by one of at most d master lists. They gave an $O(dn)$-time algorithm for verifying a proposed matching, as well as subquadratic algorithms and conditional lower bounds for related models. Finding a stable matching faster than Gale–Shapley remained open even for $d = 2$, which they suggested as the natural starting point. The two-list problem was also posed at Dagstuhl Seminar 16451, asking for a linear-time algorithm or a superlinear conditional lower bound (Lewenstein et al., 2016).

The problem. A *two-list instance* consists of sets M and W with $|M| = |W| = n$, two permutations σ_0, σ_1 of W, two permutations π_0, π_1 of M, and a *type* $t(x) \in \{0, 1\}$ for every agent $x \in M \cup W$. Man m ranks the women by $\sigma_{t(m)}$, and woman w ranks the men by $\pi_{t(w)}$. All pairs are acceptable. The input has size $O(n)$. The task is to output a stable matching.

Our result. We obtain the following deterministic bound.

Theorem 1.1. *A stable matching of a two-list instance with n men and n women can be computed deterministically in $O\big(n^2\sqrt{\log\log n / \log n}\big)$ time on a word RAM with $\Theta(\log n)$-bit words.*

The bound is $o(n^2)$, although not $O(n^{2-\varepsilon})$ for any constant $\varepsilon > 0$. The algorithm returns the unique stable matching that minimizes the number of pairs in $M_0 \times W_1$, where $M_i = \{m : t(m) = i\}$ and $W_j = \{w : t(w) = j\}$. This characterization is proved in Proposition A.3. Relabelling the types gives the same time bound for minimizing or maximizing the number of pairs in any $M_i \times W_j$.

Overview. The four groups form the cycle $M_0\, W_0\, M_1\, W_1\, M_0$, with all members of each group using the same master list. If the group graph were a path, some acceptable pair would consist of mutual first choices. Such a pair belongs to every stable matching, so repeatedly removing these pairs finds the unique stable matching (Lemma 2.1). The cycle need not contain any mutual first choices.

We remove the cycle by unrolling it. The *opened cover* $\mathcal{C}_R$ has R copies of the instance, called *sheets*, numbered $0, \ldots, R-1$. A type-0 man in sheet a is acceptable to the type-1 women in sheet $a-1$, rather than those in sheet a. All other acceptable pairs stay within a sheet. The $4R$ groups now form a path (Figure 1), so the cover has a unique stable matching ν_R.

Theorem 1.2 (informal version of Theorem 4.11)**.** *Let $R = 4n + 2$. In ν_R, the men of sheet $2n + 1$ are matched to copies of their partners in a stable matching of the original instance.*

The proof starts with a cover of $O(n^2)$ sheets, used only for analysis. Moving from left to right, every man's partner improves and every woman's partner worsens. Since each agent's partner can change only n times, two consecutive sheets are stationary for every agent. They project to a stable matching μ. Monotonicity then describes each agent's copies as a block worse than μ, a block equal to μ, and a block better than μ, in the appropriate order.

We bound these end blocks using a directed graph whose vertices are the pairs of μ. A copy of a man m who is better off with $\mu(h)$ contributes one unit of flow from h to m. Flow conservation limits the amount entering a strongly connected component from outside. Local stability inequalities let us follow predecessors backward through components of positive winding, then first successors forward through components of negative winding. This walk changes direction at most once and visits at most n vertices. A two-list separation property, proved in Lemma 4.8, bounds the flow at its terminal exit. Each end consequently has length at most $2n + 1$. Retaining the two ends and replacing the intervening sheets by the lift of μ produces a stable matching of $\mathcal{C}_{4n+2}$. Uniqueness identifies it with ν_{4n+2}.

Computing this cover one pair at a time takes $\Theta(n^2)$ time. The elimination process has a compact local description, however: the members still present in a group are exactly those not yet passed by two frontiers advancing through two fixed permutations. We pack a window of

$\Theta(\sqrt{\log n / \log\log n})$ groups, each restricted to buckets of the same size, into a machine word. A precomputed table batches the internal eliminations, yielding a Four-Russians speedup (Arlazarov et al., 1970).

Related work. The structure of stable matchings is treated in the books of Gusfield and Irving (1989) and Roth and Sotomayor (1990). Irving et al. (2008) studied stable marriage with a single master list on each side, allowing ties and incomplete lists. Cheng and Rosenbaum (2021) showed that, with incomplete lists, instances in which every list is a sublist of one of two master lists per side realize every rotation poset. Thus the family of all stable matchings can be as complex as in general instances, and counting stable matchings, which is #P-complete in general (Irving and Leather, 1986), remains hard. This concerns incomplete lists and the structure of all stable matchings, rather than the time needed to find one in the complete two-list model.

Our monotonicity lemma uses the comparative statics of adding agents to a market described by Gale and Sotomayor (1985) and Roth and Sotomayor (1990). On the lower-bound side, the linear-time verifier for two-list instances (Künnemann et al., 2019) presents an obstacle to a superlinear SETH-based lower bound via deterministic fine-grained reductions: the framework of Carmosino et al. (2016) would turn such a reduction into a refutation of the Nondeterministic Strong Exponential Time Hypothesis (NSETH).

Organization. Section 2 gives two market lemmas. Section 3 introduces the cover and its stationary slice. Section 4 bounds the two ends and proves the structural theorem. Section 5 gives the fast implementation, and Section 6 discusses further directions. Section A gives a count-matrix characterization and an exact gap identity that identify the matching returned by the algorithm.

2 Preliminaries

Markets and stability. We use general two-sided markets in which every agent finds only some agents of the other side acceptable, and a pair is acceptable if each of its agents finds the other acceptable. Every agent has a strict ranking of its acceptable partners. Being unmatched, written $\perp$, is worse than any acceptable partner. A matching uses acceptable pairs only, and it is *stable* if no acceptable pair (x, y) consists of two agents who both prefer each other to their partners. We write $y \succ_x y'$ if x prefers y to y', and $y \succeq_x y'$ if $y \succ_x y'$ or $y = y'$.

In a two-list instance all pairs are acceptable. We write $M_i = \{m : t(m) = i\}$ and $W_j = \{w : t(w) = j\}$. The pairs in $M_0 \times W_1$ play a special role. We call them *seam pairs* and set

$$\kappa(m, w) = \begin{cases} 1 & \text{if } t(m) = 0 \text{ and } t(w) = 1, \\ 0 & \text{otherwise,} \end{cases} \qquad q(\mu) = \sum_{m \in M} \kappa(m, \mu(m))$$

for every perfect matching μ. Thus $q(\mu)$ is the number of seam pairs in μ.

Group markets. A *group market* is given by a partition of the agents into *groups*, each consisting of agents of one side, and a bipartite graph on the groups, the *group graph*, with the following properties.

- Every group G has a strict order $\prec_G$ on the agents of the groups adjacent to G (smaller is better), and every agent of G ranks its acceptable partners by $\prec_G$.
- Every agent x of G *accepts* an initial segment of $\prec_G$.
- Two agents form an acceptable pair if and only if each accepts the other.

In particular, only agents of adjacent groups can form acceptable pairs. Two agents are *mutual first choices* if they form an acceptable pair and each is the other's most preferred acceptable partner. Such a pair belongs to every stable matching, since otherwise it would block it.

Lemma 2.1 (Forest lemma)**.** *Let a group market have a group graph that is a forest. If at least one acceptable pair exists, then some acceptable pair consists of mutual first choices. Consequently the market has exactly one stable matching, and it is obtained by repeatedly matching a pair of mutual first choices and removing both agents, until no acceptable pair remains.*

Proof. Call a group *live* if one of its agents belongs to an acceptable pair. For a live group G, let h_G be the $\prec_G$-first agent among all agents that form an acceptable pair with some member of G, and let $\varphi(G)$ be the group of h_G. Then $\varphi(G)$ is live and adjacent to G. Starting from any live group and applying φ repeatedly, we eventually revisit a group, which gives distinct groups $G_1, \dots, G_k$ with $\varphi(G_i) = G_{i+1}$ for all i (indices modulo k). Here $k \geq 2$, since $\varphi(G)$ lies on the other side than G. If $k \geq 3$, the groups $G_1, \dots, G_k$ would form a cycle in the group graph. Hence $k = 2$: there are adjacent groups G and H with $h_G \in H$ and $h_H \in G$.

We claim that h_G and h_H are mutual first choices. By definition, h_G forms an acceptable pair with some $x \in G$. Since x forms an acceptable pair with a member of H, namely h_G, the definition of h_H gives $h_H \preceq_H x$. Now h_G accepts x, and it accepts an initial segment of $\prec_H$, so it also accepts h_H. Symmetrically, h_H accepts h_G, so the two form an acceptable pair. Every acceptable partner y of h_H forms an acceptable pair with a member of G, so $h_G \preceq_G y$. Hence h_G is the first choice of h_H, and symmetrically h_H is the first choice of h_G.

For the second statement, let (x, y) be mutual first choices, and let N' be the market without x and y. It is again a group market with the same group graph. Every stable matching contains (x, y), and removing this pair from it leaves a stable matching of N'. Conversely, adding (x, y) to a stable matching of N' gives a stable matching, because x and y get their first choices. By induction on the number of agents, N' has exactly one stable matching, obtained by the stated procedure, and hence so does the original market. If no acceptable pair exists, the empty matching is the only matching. □

The second lemma is the classical effect of adding agents of one side to a market described by Gale and Sotomayor (1985) and Roth and Sotomayor (1990): the other side gains and the same side loses. We need it in a form where the new agents may include agents of both sides, as long as only agents of one side interact with the old market, so we include the short proof.

Lemma 2.2 (Separated entry)**.** *Let N be a market and O a subset of its agents. Suppose every acceptable pair with one agent in O and one agent outside O consists of a man in O and a woman outside O. If the submarket on O and the market N have unique stable matchings ν_O and ν_N, then every man in O weakly prefers ν_N to ν_O, and every woman in O weakly prefers ν_O to ν_N. The same holds with the roles of men and women exchanged.*

Proof. Run the men-proposing deferred acceptance algorithm on N. Its output is stable, hence equal to ν_N. Suppose that some man of O is at some point rejected by his partner in ν_O, and consider the first time this happens: $m \in O$ is rejected by $w = \nu_O(m)$. Then $w \in O$ holds a man m' whom she prefers to m. Only men of O form acceptable pairs with w, so $m' \in O$. If m' is unmatched in ν_O, he prefers w to $\perp$. Otherwise, he has not yet been rejected by $\nu_O(m')$, and proposals follow his preference order, so he prefers w to $\nu_O(m')$ (note $w \neq \nu_O(m')$). Now (m', w) is an acceptable pair of O that blocks ν_O, a contradiction. Hence every man of O who is matched in ν_O ends with $\nu_O(m)$ or someone better. Unmatched men of O have nothing to lose.

Suppose a woman $w \in O$ strictly prefers $m' = \nu_N(w)$ to $\nu_O(w)$. Again $m' \in O$, and by the first part m' weakly prefers w to $\nu_O(m')$. Since $w \neq \nu_O(m')$, the preference is strict, so (m', w) blocks ν_O, a contradiction. The last sentence follows by symmetry, using the women-proposing algorithm. □

3 The opened cover

Fix a two-list instance I with $n \geq 1$. Throughout, $[R] = \{0, 1, \ldots, R-1\}$.

Definition 3.1 (Opened cover)**.** For an integer $R \geq 1$, the *opened cover* $\mathcal{C}_R$ is the market with a man m^z and a woman w^z for every $m \in M$, $w \in W$ and $z \in [R]$. The copies m^a and w^b form an acceptable pair if and only if

$$b = a - \kappa(m, w).$$

Every copy ranks its acceptable partners as the original agent ranks their originals. We call z the *sheet* of m^z and w^z.

So a type-0 man of sheet a is acceptable to the type-0 women of sheet a and to the type-1 women of sheet $a-1$, while a type-1 man of sheet a is acceptable to all women of sheet a. Let M_i^z and W_j^z denote the copies in sheet z of the groups M_i and W_j. The acceptable pairs are exactly the pairs between consecutive groups of the sequence

$$M_0^0,\ W_0^0,\ M_1^0,\ W_1^0,\ M_0^1,\ W_0^1,\ M_1^1,\ W_1^1,\ \ldots,\ M_1^{R-1},\ W_1^{R-1}, \tag{3.1}$$

and between two consecutive groups every pair is acceptable (Figure 1). All copies in a group rank by the same master list, and the two groups adjacent to a group contain at most one copy of each original agent, so this list orders them. Hence $\mathcal{C}_R$ is a group market whose group graph is a path, in which every agent accepts all agents of the adjacent groups, and Lemma 2.1 gives:

Corollary 3.2. *$\mathcal{C}_R$ has exactly one stable matching ν_R. It can be computed in $O(nR)$ time by repeatedly matching mutual first choices.*

Proof. Everything except the running time is Lemma 2.1. For the running time, keep the members of each group in two linked lists, sorted by the master lists of its two neighbouring groups. Every member of a group has the same first choice, namely the better of the first members of the two neighbouring groups' lists, so mutual first choices can be found by inspecting the heads of these lists. Each removal takes $O(1)$ time and affects the first choices of $O(1)$ groups. Section 5.1 gives the details. □

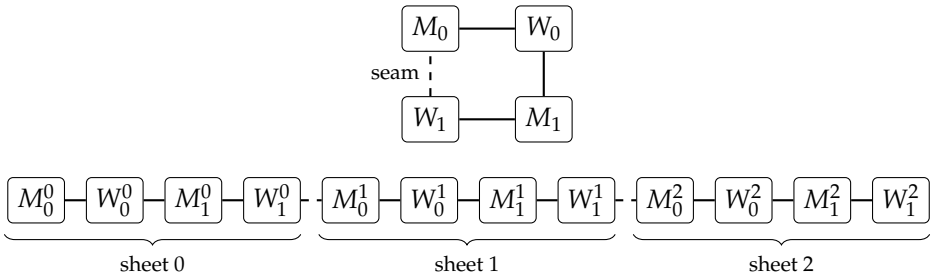


Figure 1: Top: the four groups of a two-list instance. Every man is acceptable to every woman, so the groups form a cycle. Bottom: the opened cover $\mathcal{C}_3$. Every edge joins two groups whose members are all mutually acceptable. The dashed edges are the copies of the seam M_0W_1: the type-0 men of sheet z and the type-1 women of sheet $z-1$ are mutually acceptable.

Every copy m^a has exactly one acceptable copy of each woman w, namely $w^{a-\kappa(m,w)}$, except that the type-0 men of sheet 0 have no acceptable type-1 women. Symmetrically, the type-1 women of sheet $R-1$ have no acceptable type-0 men. Letting the sheet index range over $\mathbb{Z}$ gives an infinite market in which every stable matching μ of I lifts to the stable perfect matching $m^z \mapsto \mu(m)^{z-\kappa(m,\mu(m))}$. The opened cover is a finite window of this infinite market, and its two ends are the only reason why ν_R is not simply a lift.

Notation. For a copy x^z, we write $\nu_R(x^z)$ for the *original* agent whose copy is matched to x^z, or $\perp$ if x^z is unmatched. Such partners are compared by the preferences of x in I, with $\perp$ last.

3.1 Monotone sheets

Lemma 3.3 (Monotone sheets)**.** *For every $R \geq 1$, man m, woman w and $0 \leq z < R-1$,*

$$\nu_R(m^{z+1}) \succeq_m \nu_R(m^z) \qquad \text{and} \qquad \nu_R(w^z) \succeq_w \nu_R(w^{z+1}).$$

Moreover, $\nu_R(m^z) \succeq_m \nu_{R+1}(m^z)$ and $\nu_{R+1}(m^{z+1}) \succeq_m \nu_R(m^z)$ for all $z \in [R]$.

Proof. Consider $\mathcal{C}_{R+1}$. Its sheets $0, \ldots, R-1$ form a copy of $\mathcal{C}_R$. The only acceptable pairs between these sheets and sheet R join type-0 men of sheet R to type-1 women of sheet $R-1$. By Lemma 2.2 (with the roles of the sexes exchanged), every man in sheets $0, \ldots, R-1$ weakly prefers ν_R to ν_{R+1}, which is the first inequality of the second statement.

Similarly, sheets $1, \ldots, R$ of $\mathcal{C}_{R+1}$ form a copy of $\mathcal{C}_R$ shifted by one sheet, and the only acceptable pairs between them and sheet 0 join type-0 men of sheet 1 to type-1 women of sheet 0. By Lemma 2.2, the men of sheets $1, \ldots, R$ weakly prefer ν_{R+1} to the shifted copy of ν_R, which is the second inequality.

Together, $\nu_{R+1}(m^{z+1}) \succeq_m \nu_R(m^z) \succeq_m \nu_{R+1}(m^z)$, so men's partners are monotone in every cover with at least two sheets. Covers with one sheet are trivial. The statement for women follows in the same way from the women's part of Lemma 2.2. □

3.2 A stationary slice

Lemma 3.4 (Stationary sheets)**.** *Suppose that $z < R-1$ and $\nu_R(x^z) = \nu_R(x^{z+1})$ for every original agent x. Then $\mu(m) = \nu_R(m^z)$ defines a stable perfect matching of the original instance, and $\nu_R(w^z) = \nu_R(w^{z+1}) = \mu^{-1}(w)$ for every woman w.*

Proof. First allow $\mu(m) = \perp$. If $\mu(m) = w$, then m^{z+1} is matched to a copy of w in sheet z or $z+1$. Stationarity of w gives $\nu_R(w^z) = m$. Conversely, if $\nu_R(w^z) = m$, her partner lies in sheet z or $z+1$, and stationarity of m gives $\mu(m) = w$. Thus these assignments are inverse partial matchings of the original agents.

If the matching is not perfect, there are an unmatched original man m and woman w. Stationarity makes the acceptable copies m^{z+1} and $w^{z+1-\kappa(m,w)}$ both unmatched, contradicting stability of ν_R. Finally, a blocking pair (m, w) of μ would lift to a blocking pair between these same two sheets. Hence μ is stable. □

Fix, for the analysis, the depth

$$L = 2n^2 + 4n + 4. \tag{3.2}$$

By Lemma 3.3, each original agent's partner changes at most n times along the sheets. At most $2n^2$ boundaries therefore see a change for any agent, so $\mathcal{C}_L$ has a pair of stationary sheets. Fix one such pair $z, z+1$, and let μ be its stable matching from Lemma 3.4. The algorithm will not construct this long cover.

Write $\kappa_m = \kappa(m, \mu(m))$ and $q = q(\mu)$. For a man m and woman w, let

$$\begin{aligned} u_m &= \big|\{a \in [L] : \nu_L(m^a) \succ_m \mu(m)\}\big|, \\ v_w &= \big|\{a \in [L] : \nu_L(w^a) \succ_w \mu^{-1}(w)\}\big|, \\ e_w &= \big|\{a \in [L] : \nu_L(w^a) = \perp\}\big|. \end{aligned}$$

Thus u_m and v_w count the copies that are better off than in μ, and e_w counts the unmatched copies of w. Set

$$U = \max_{m \in M} u_m, \qquad V = \max_{w \in W} v_w.$$

All these quantities refer to the fixed cover $\mathcal{C}_L$.

Lemma 3.5 (Partner blocks). *Every used pair of originals (m, w) with $w \neq \mu(m)$ makes exactly one endpoint better off than in μ. For each man m, the copies matched to $\mu(m)$ are precisely those with*

$$v_{\mu(m)} + \kappa_m \leq a < L - u_m. \tag{3.3}$$

For each woman w, writing $h = \mu^{-1}(w)$, the copies matched to h are precisely those with

$$v_w \leq a < L - u_h - \kappa_h. \tag{3.4}$$

In particular, the pair $(m, \mu(m))$ is used $L - u_m - v_{\mu(m)} - \kappa_m$ times.

Proof. Both endpoints cannot be better off, since μ is stable. Suppose both were worse off, and let m^a be matched to $w^{a-\kappa(m,w)}$. Since m has his μ-partner in sheet z, monotonicity gives $a < z$. The woman's sheet is also before z, where she is at least as well off as with her μ-partner, a contradiction.

By monotonicity, m is better off in exactly his last u_m sheets, and $w = \mu(m)$ is better off in exactly her first v_w sheets. Their nonempty blocks of μ-partners have equal length, say c_m. The man's block ends at $L - u_m$, and the woman's block begins at v_w. Every matching edge shifts the sheet index by κ_m, so the man's block begins at $v_w + \kappa_m$. This proves both intervals and the formula for c_m. □

The copies before a man's block in (3.3) are worse off or unmatched. Those after it are better off. For a woman the order is reversed. We call these outer blocks the *left* and *right ends* of the cover. Their shape is shown in Figure 2.

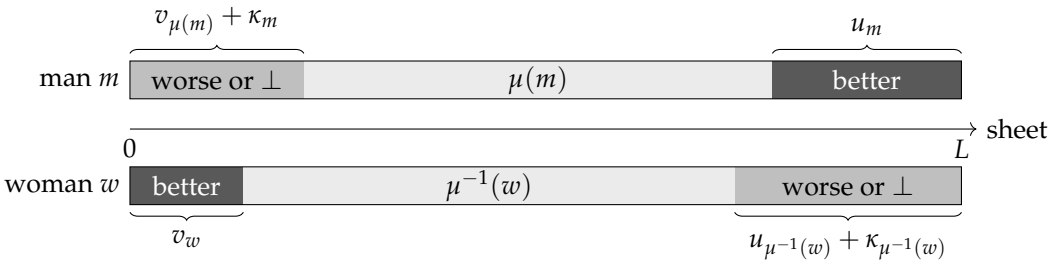


Figure 2: The partner blocks in the fixed long cover (Lemma 3.5). The outer blocks may be empty. By Theorem 4.9, their lengths are at most $2n + 1$.

4 The ends are short

We first prove $U, V \leq q + n$ for the fixed long cover, and then splice its two ends into a cover of $4n + 2$ sheets. The flow argument uses only the partner blocks of Lemma 3.5.

4.1 The flow at the right end

Contract each pair $(h, \mu(h))$ to a vertex named h. The *willingness graph* G has vertex set M and an arc $h \to m$, for $h \neq m$, whenever

$$\mu(h) \succ_m \mu(m).$$

Give this arc flow

$$f(h \to m) = \big|\{a \in [L] : \nu_L(m^a) = \mu(h)\}\big|.$$

The graph includes arcs with zero flow. An arc means that m would prefer $\mu(h)$ to his own partner. Stability of μ then makes $\mu(h)$ prefer h to m. By Lemma 3.5, the incoming flow at m counts exactly his better-off copies, while the outgoing flow at h counts the copies of $\mu(h)$ that are matched to worse partners. Thus

$$\mathrm{in}(m) = \sum_{h \to m} f(h \to m) = u_m.$$

Lemma 4.1 (Conservation)**.** *For every vertex h,*

$$\mathrm{out}(h) - \mathrm{in}(h) = \kappa_h - e_{\mu(h)}. \tag{4.1}$$

Consequently, ν_L leaves exactly q men and q women unmatched.

Proof. The final $u_h + \kappa_h$ copies of $\mu(h)$ are worse off or unmatched. Of these, $e_{\mu(h)}$ are unmatched, and all the others are counted by the outgoing flow. Hence $\mathrm{out}(h) = u_h + \kappa_h - e_{\mu(h)}$. Summing over h cancels the incoming and outgoing flows and gives $\sum_w e_w = \sum_h \kappa_h = q$. The two sides of the cover have equal size, so the number of unmatched men is also q. □

We refer to the strongly connected components of G simply as *components*.

Lemma 4.2 (External inflow)**.** *The total flow entering any component from outside is at most q.*

Proof. By (4.1), the sum of the positive excesses $\mathrm{out}(h) - \mathrm{in}(h)$ is at most $\sum_h \kappa_h = q$. Decompose the nonnegative flow into cycle flows and path flows from vertices of positive excess to vertices of negative excess. Such a decomposition can be obtained by removing cycles and then repeatedly following a positive-flow arc from a source and subtracting the minimum flow along the resulting path. The total weight of the paths is the total positive excess, at most q.

Every cycle stays in one component. A directed path cannot leave a component and later re-enter it, since the component graph is acyclic. Each path therefore contributes at most its weight to the flow entering a given component. □

For an arc $h \to m$, set

$$I_m(h) = \sum_{\substack{h' \to m \\ \mu(h') \succ_m \mu(h)}} f(h' \to m),$$

$$O_h(m) = \sum_{\substack{h \to m' \\ m' \succ_{\mu(h)} m}} f(h \to m').$$

The first sum counts incoming flow from women whom m prefers to $\mu(h)$. The second counts outgoing flow to men whom $\mu(h)$ prefers to m. Define

$$\widetilde{u}_h = u_h + \kappa_h, \qquad \omega(h \to m) = \kappa(m, \mu(h)) - \kappa_m \in \{-1, 0, 1\}. \tag{4.2}$$

Lemma 4.3 (Local inequality)**.** *For every arc $h \to m$,*

$$\widetilde{u}_h \le I_m(h) + f(h \to m) + O_h(m) + \kappa(m, \mu(h)), \tag{4.3}$$

with equality whenever $f(h \to m) > 0$.

Proof. Write $w = \mu(h)$, $A = I_m(h) + f(h \to m)$, and

$$B = L - \widetilde{u}_h + O_h(m).$$

The copies of m with a partner at least as good as w occupy exactly the last A sheets. By Lemma 3.5, the copies of w with a partner strictly better than m consist of those with a partner at least as good as h, followed by those outgoing-flow partners she prefers to m. They occupy exactly the first B sheets.

If $B + \kappa(m, w) < L - A$, choose the integer $a = B + \kappa(m, w)$. Then $\kappa(m, w) \le a < L$, the partner of m^a is worse than w, and the partner of $w^{a-\kappa(m,w)}$ is not strictly better than m. It cannot be a copy of m, since the only such acceptable copy is m^a, which is not matched to her. These copies therefore block v_L. Stability gives $B + \kappa(m, w) \ge L - A$, which is (4.3).

If $f(h \to m) > 0$, the copies of m matched to w begin at sheet $L - A$, and the copies of w matched to m begin at sheet B. Their intervals have the same positive length and are shifted by $\kappa(m, w)$, so $L - A = B + \kappa(m, w)$. This gives equality. □

We use the local inequality in two directions. If h has an outgoing arc, let its *first successor* $F(h)$ be the out-neighbour whom $\mu(h)$ likes best, whether or not that arc has positive flow. Since $O_h(F(h)) = 0$ and $I_m(h) + f(h \to m) \le u_m$, we have

$$\widetilde{u}_h \le \widetilde{u}_{F(h)} + \omega(h \to F(h)). \tag{F}$$

If $u_m > 0$, let $P(m)$ supply the woman whom m likes least among his positive-flow incoming arcs. Then $I_m(P(m)) + f(P(m) \to m) = u_m$. Equality in (4.3) gives

$$\widetilde{u}_m + O_{P(m)}(m) = \widetilde{u}_{P(m)} - \omega(P(m) \to m),$$

and hence

$$\widetilde{u}_m \le \widetilde{u}_{P(m)} - \omega(P(m) \to m). \tag{P}$$

Each move costs at most one in the bound on $\widetilde{u}$. Winding will determine which direction to follow.

4.2 Winding

The *winding* of a directed walk is the sum of the ω-values of its arcs. Closed walks below are always nonempty.

Lemma 4.4 (No zero winding). *No closed walk in G has winding 0.*

Proof. Consider the infinite cover $\mathcal{C}_{\mathbb{Z}}$ (Definition 3.1 with sheets in $\mathbb{Z}$) and the lift $\tilde{\mu}$ of μ, which matches h^a with $\mu(h)^{a-\kappa_h}$. Let $h_0 \to h_1 \to \cdots \to h_k = h_0$ be a closed walk of winding 0. Put $z_0 = 0$ and $z_{i+1} = z_i + \omega(h_i \to h_{i+1})$, so that $z_k = z_0$. Let $x_i = \mu(h_i)^{z_i}$ and $y_i = h_i^{z_i+\kappa_{h_i}}$, so that $\tilde{\mu}$ matches y_i with x_i. By the definition of ω, the copy x_i is acceptable to y_{i+1}, and h_{i+1} prefers $\mu(h_i)$ to $\mu(h_{i+1})$. So every man copy y_{i+1} prefers a woman copy of the finite set $X = \{x_0, \dots, x_{k-1}\}$ to his $\tilde{\mu}$-partner.

Now consider the finite market on $X \cup Y$, where $Y = \{y_0, \dots, y_{k-1}\}$. Each $y \in Y$ accepts exactly the acceptable copies in X that he likes at least as much as his $\tilde{\mu}$-partner, and each $x \in X$ accepts all acceptable copies in Y. This is a group market: its groups are the intersections of $X \cup Y$ with the groups of $\mathcal{C}_{\mathbb{Z}}$, and every agent accepts an initial segment of its list. Its group graph is a finite subgraph of a path, hence a forest. It contains the acceptable pairs (y_{i+1}, x_i). Every man's first choice is strictly better than his $\tilde{\mu}$-partner. Every man $y \in Y$ that accepts $x \in X$, other than $\tilde{\mu}(x)$, prefers x to his own partner, so by stability of $\tilde{\mu}$ the woman x prefers $\tilde{\mu}(x) \in Y$ to y. Hence every woman's first choice is her $\tilde{\mu}$-partner. So there is no pair of mutual first choices, contradicting Lemma 2.1. □

A component is *trivial* if it consists of a single vertex. Since G has no loops, a trivial component contains no closed walk.

Corollary 4.5. *Every nontrivial component of G is either* positive, *meaning that all its closed walks have positive winding, or* negative, *meaning that all its closed walks have negative winding.*

Proof. Fix a vertex x of the component. Let D be any closed walk in the component, let P be a walk from x to a vertex of D, and let P' be a walk from that vertex back to x. Going along P, around D exactly j times, and back along P' gives a closed walk through x of winding $\omega(P) + \omega(P') + j\,\omega(D)$. By Lemma 4.4, $\omega(D) \neq 0$, so for large j this winding has the same sign as $\omega(D)$. Hence, if the component contained closed walks of both signs, it would contain closed walks through x of windings $a > 0$ and $-b < 0$. Going b times around the first and a times around the second gives a closed walk of winding 0, contradicting Lemma 4.4. □

For two men x, y and $i \in \{0, 1\}$, we write $x <_i y$ if σ_i ranks $\mu(x)$ before $\mu(y)$. Thus

$$h \to m \qquad \Longleftrightarrow \qquad h <_{t(m)} m.$$

Lemma 4.6 (Cycles and two-cycles). *(i) Every cycle of G has vertices of both types.*

(ii) If $h \to m \to h$ is a cycle of length two, then $t(h) \neq t(m)$ and $t(\mu(h)) \neq t(\mu(m))$. Its winding is positive if $t(h) = t(\mu(h))$ and negative otherwise.

Proof. (i) Along a cycle all of whose vertices have type i, every arc $x \to y$ means $x <_i y$, which is impossible around a cycle. (ii) Since $\kappa(m, w) = (1 - t(m))\,t(w)$, we have $\omega(h \to m) = (1 - t(m))(t(\mu(h)) - t(\mu(m)))$, and therefore

$$\omega(h \to m) + \omega(m \to h) = \big(t(h) - t(m)\big)\big(t(\mu(h)) - t(\mu(m))\big).$$

By (i) $t(h) \neq t(m)$, and by Lemma 4.4 the product is nonzero, so $t(\mu(h)) \neq t(\mu(m))$. The product is positive exactly when the two differences have the same sign, that is, when $t(h) = t(\mu(h))$. □

4.3 First successors and shielding

The next observation identifies two useful vertices on two-cycles. Write $x \leq_i y$ when $x <_i y$ or $x = y$.

Lemma 4.7 (Extremal vertices). *Let S be a nontrivial component and $i \in \{0, 1\}$.*

(i) The first vertex of S in $<_i$ has type $1 - i$ and lies on a two-cycle.
(ii) The last type-i vertex of S in $<_i$ lies on a two-cycle.

Proof. For (i), let p be the first vertex of S in $<_i$. If $t(p) = i$, an internal in-neighbour of p would precede it in $<_i$, a contradiction. Hence $t(p) = 1 - i$. Take a directed cycle through p, and let p' be its last type-i vertex before returning to p. Such a vertex exists by Lemma 4.6. Every subsequent vertex up to p has type $1 - i$, so transitivity gives $p' <_{1-i} p$, and thus $p' \to p$. Minimality gives $p <_i p'$, so $p \to p'$ as well.

For (ii), let r be the last type-i vertex in $<_i$. It has no internal out-neighbour of type i. On a cycle beginning at r, let x be the first subsequent type-i vertex, and let r' precede x. Between r and x there is a nonempty sequence of type-$1 - i$ vertices. Transitivity gives $r <_{1-i} r'$, while $r' <_i x \leq_i r$. Hence $r \to r' \to r$ is a two-cycle. □

Lemma 4.8 (Shielding). *Let S be a negative component and T a positive component. If $h \in S$ and $m = F(h) \in T$, then m prefers $\mu(h)$ to $\mu(x)$ for every $x \in T$.*

Proof. Put $i = t(m)$ and $j = 1 - i$. There is an arc from S to T, so there is no arc from T back to S. Equivalently,

$$y <_{t(y)} x \qquad \text{for every } y \in S \text{ and } x \in T. \tag{4.4}$$

Suppose, contrary to the claim, that $x_0 <_i h$ for some $x_0 \in T$. Then $t(h) = j$, since otherwise (4.4) gives $h <_i x_0$. Moreover, every internal out-neighbour y of h has type j. If $t(y) = i$, then

$$h <_i y <_i x_0 <_i h,$$

using the arc, (4.4), and the assumption.

Let r be the last type-j vertex of S in $<_j$. Since h has an internal out-neighbour, $h <_j r$ and $h \to r$. By Lemma 4.7, r lies on a two-cycle $r \to r' \to r$, with $t(r') = i$. This cycle is negative, so Lemma 4.6 gives $t(\mu(r)) = i$. Also,

$$r <_i r' <_i x_0 <_i h <_i m,$$

where the second inequality is (4.4). Thus $r \to m$, and stability of μ makes $\mu(r)$ prefer r to m. The choice $F(h) = m$, on the other hand, makes $\mu(h)$ prefer m to r. These two women rank m, r oppositely, so they use different lists. It follows that $t(\mu(h)) = j$.

Now let p be the first vertex of T in $<_i$. By Lemma 4.7, p has type j and lies on a two-cycle. Positivity of T gives $t(\mu(p)) = j$. Since

$$p \leq_i x_0 <_i h <_i m,$$

we have $p \to m$, and stability makes $\mu(p)$ prefer p to m. But (4.4) gives $h <_j p$, hence $h \to p$, so the choice $F(h) = m$ makes $\mu(h)$ prefer m to p. The women $\mu(h)$ and $\mu(p)$ have the same type j, yet rank m, p oppositely, a contradiction. □

4.4 Bounding the ends

Theorem 4.9 (Short ends). *For the fixed cover $\mathcal{C}_L$, we have $U \leq q + n$ and $V \leq q + n$.*

Proof. Let N^- be the vertices in negative components and $N^\circ = M \setminus N^-$ those in positive or trivial components. Starting at any vertex v, construct the following walk. While the current vertex m is in N° and $u_m > 0$, move backward to $P(m)$, using (P). Stop if a vertex of zero inflow is reached. If the walk reaches N^-, change direction and follow F forward for as long as the current vertex remains in N^-. We stop at the last negative vertex h, before its successor $F(h)$ leaves N^-.

There is no repetition in the backward part. A repetition would give a directed closed walk in a single component of N°, necessarily a positive component. Summing (P) around that walk would make its winding nonpositive. Similarly, there is no repetition in the forward part: summing (F) around a repetition in a negative component would make its winding nonnegative. Both conclusions contradict Corollary 4.5. Every negative vertex has an outgoing arc, so the forward part must eventually exit. The two parts use disjoint sets of vertices. Excluding the terminal exit, the walk therefore visits at most n vertices.

If it ends at a vertex h with zero inflow, then $\widetilde{u}_h = \kappa_h \leq 1$. Otherwise let $m = F(h) \notin N^-$, and let T be the component of m. If T is trivial, all its incoming flow is external. If T is positive, Lemma 4.8 says that every wife whom m likes at least as much as $\mu(h)$ belongs to a vertex outside T. In either case, Lemma 4.2 gives

$$I_m(h) + f(h \to m) \leq q.$$

Since $O_h(m) = 0$, (4.3) yields

$$\widetilde{u}_h \leq q + \kappa(m, \mu(h)) \leq q + 1.$$

Each preceding move costs at most one by (F) or (P), and there are at most $n - 1$ such moves. Hence $u_v \leq \widetilde{u}_v \leq q + n$ for every v, proving $U \leq q + n$.

For the other bound, exchange men and women, and complement every type. In the resulting instance I', define

$$t'(x) = 1 - t(x) \qquad (x \in M \cup W), \qquad \sigma'_i = \pi_{1-i}, \quad \pi'_j = \sigma_{1-j}.$$

All preferences are unchanged and $\kappa'(w, m) = \kappa(m, w)$. Reversing the sheets, $a \mapsto L - 1 - a$, takes the acceptance rule of $\mathcal{C}_L(I)$ to that of $\mathcal{C}_L(I')$. Uniqueness therefore identifies the reversed matching with the stable matching of the new cover. The two stationary sheets remain stationary, with matching μ viewed from the other side. Its seam count is still q, and the better-off copies of a new man w are exactly the v_w better-off copies of the original woman. Applying the established right-end bound gives $V \leq q + n$. □

4.5 Splicing the cover

The long cover can now be shortened without changing its two ends. This step uses the orientation of the seam: only women on the left of a cut can be matched to men on its right.

Lemma 4.10 (Splicing)**.** *For every $R \geq U + V + 2$, the stable matching of $\mathcal{C}_R$ agrees with the lift of μ on every man in a sheet a satisfying*

$$V + 1 \leq a \leq R - U - 1. \tag{4.5}$$

Proof. Retain from $\mathcal{C}_L$ its first $V + 1$ sheets and its last $U + 1$ sheets. These pieces are disjoint, since Theorem 4.9 gives $U, V \leq 2n$ and (3.2) gives $L > 4n + 2$. Every woman in the last sheet V of the left piece has her μ-partner by (3.4). Every man in the first sheet $L - U - 1$ of the right piece has his μ-partner by (3.3). Thus all matching edges cut at these two boundaries are seam edges of the lift of μ.

Place $R - U - V - 2$ new sheets between the retained pieces, and renumber the resulting sheets from 0 to $R - 1$. Keep the matching edges internal to each retained piece, use the lift of μ on the inserted sheets, and reconnect the severed seam edges according to μ. This is a well-defined matching. At each new seam, the severed endpoints correspond to exactly the pairs $(m, \mu(m))$ with $\kappa_m = 1$. Reconnecting those pairs matches all such endpoints once. Every agent in a retained piece keeps the same original partner, including the possibility of being unmatched.

An acceptable pair internal to a retained piece is nonblocking because the original matching was stable and both original partners are unchanged. An acceptable pair internal to the inserted sheets is nonblocking by stability of μ. Every other acceptable pair crosses a new seam. Its endpoints are a woman in the last sheet on the left and a man in the first sheet on the right, and both have their μ-partners. Stability of μ again rules out a block. This includes the case of no inserted sheets, when the retained pieces are joined directly.

The constructed matching is therefore stable in $\mathcal{C}_R$, and equals ν_R by Corollary 3.2. The inserted sheets, together with the first sheet of the right piece, give exactly the men's range in (4.5). □

Theorem 4.11 (Structural theorem)**.** *Let $R = 4n + 2$ and $a = 2n + 1$. In the unique stable matching ν_R of $\mathcal{C}_R$, every man m^a is matched to $\mu(m)^{a-\kappa_m}$. In particular, $m \mapsto \nu_R(m^a)$ is a stable matching of the original instance.*

Proof. By Theorem 4.9, $U, V \leq q + n \leq 2n$. Thus $R \geq U + V + 2$ and $V + 1 \leq 2n + 1 \leq R - U - 1$. Apply Lemma 4.10. □

5 The algorithm

By Theorem 4.11, it suffices to compute the stable matching ν_R of $\mathcal{C}_R$ for $R = 4n + 2$, in an implicit form that reveals the partners of the men of sheet $2n + 1$. Doing so by matching mutual first choices one pair at a time (Corollary 3.2) takes $\Theta(n^2)$ time. We now show how to save a factor of $\Theta(\sqrt{\log n / \log \log n})$.

5.1 Elimination with frontiers

Number the groups of (3.1) as $G_1, \dots, G_K$, where $K = 4R$. Each G_k is a copy of M_0, W_0, M_1 or W_1, called its *kind*, and its members rank the other side by the list of their kind, which we call the *list of* G_k. Every member of G_k has a *left position*, its position in the list of G_{k-1}, and a *right position*, its position in the list of G_{k+1}. Both lists are permutations of all n agents of the side of G_k, and positions range over $0, \dots, n-1$. For example, the left and right positions of the members of M_0^z are their positions in π_1 (the list of W_1^{z-1}) and in π_0 (the list of W_0^z). For G_1 and G_K, use the master list prescribed by the periodic four-group pattern for the missing neighbour, but treat that neighbour itself as an empty group.

We run the elimination of Lemma 2.1. At any time, the *left head* of G_k is its present member with the smallest left position, and the *right head* is its present member with the smallest right position. A missing head is ignored. If both neighbouring groups are empty, there is no favourite. Otherwise, all members of G_k rank the present members of $G_{k-1} \cup G_{k+1}$ by the list of G_k, so they all have the same first choice, the *favourite* of G_k: the better, in the list of G_k, of the right head of G_{k-1} and the left head of G_{k+1}. Nonempty adjacent groups G_k, G_{k+1} are *ready* if the favourite of G_k is the left head of G_{k+1} and the favourite of G_{k+1} is the right head of G_k. Then these two heads are mutual first choices, and every pair of mutual first choices arises in this way. By Lemma 2.1, we may match the heads of ready pairs in any order, until no pair is ready.

Let the *frontiers* λ_k and ρ_k be the left and right positions of the left and right heads of G_k, and n if G_k is empty. A member removed through the left side was the left head at that moment, so its left position is smaller than every later value of λ_k. The same holds on the right. Hence both frontiers only move forward, and we have the following invariant:

$$\begin{gathered}\text{the present members of } G_k \text{ are exactly its members with} \\ \text{left position at least } \lambda_k \text{ and right position at least } \rho_k.\end{gathered} \tag{5.1}$$

Lemma 5.1 (Order-preserving edges)**.** *For adjacent groups G_k, G_{k+1}, let $g_1, g_2, \dots$ be the members of G_k matched into G_{k+1}, sorted by right position, and let $g'_1, g'_2, \dots$ be the members of G_{k+1} matched into G_k, sorted by left position. Then g_i is matched with g'_i for every i.*

Proof. Each pair across this edge consists of the right head of G_k and the left head of G_{k+1} at the time it is created. Since both heads move forward, the i-th pair created consists of g_i and g'_i. □

So the matching across an edge is determined by two *chosen vectors*: the n-bit vector that marks the right positions of the members of G_k matched into G_{k+1}, and the one that marks the left positions of the members of G_{k+1} matched into G_k. The i-th marked member on one side is matched to the i-th marked member on the other. For $z = 2n + 1$, the chosen vectors of the four edges incident to M_0^z and M_1^z give the partners of all men of sheet z in $O(n)$ time.

5.2 Packing

The implementation stores the shared permutations and compressed group states, rather than materializing all copies of the agents. We handle bounded n directly by Gale–Shapley. For

sufficiently large n, let $B \geq 2$ and $s \geq 3$ be integer parameters, chosen in Section 5.3. We divide the positions $0, \ldots, n-1$ of every list into *buckets* of B consecutive positions. For a group and a side, the *current bucket* is the bucket containing the frontier. By (5.1), no present member has a position in an earlier bucket.

Local state. The local state of a group G_k consists of

- for each side, an *active mask* of B bits, marking the positions in the current bucket that hold present members of G_k, and a *chosen mask* of B bits, marking the positions in the current bucket of members removed through this side.
- the *descriptor* of its two current buckets (see below).
- a comparison of the bucket indices, in the list of G_k, of the right head of G_{k-1} and the left head of G_{k+1} (smaller, equal or larger).
- a flag that is set when G_k is *empty*, that is, when all its members have been removed, and two flags that record whether G_{k-1} and G_{k+1} are empty.

The bucket indices themselves are kept in ordinary arrays. Only their comparisons enter the packed state. Every group also stores its fixed kind, using two bits. Padded positions in the last bucket never hold an agent. A head is the first marked position of the corresponding active mask. Hence the local states of $G_{k-1}, \ldots, G_{k+2}$ determine whether G_k and G_{k+1} are ready: two heads are compared by their bucket comparison and, if their buckets are equal, by their offsets.

Descriptors. Fix a kind and a bucket β of the left positions and a bucket β' of the right positions of its members. The descriptor of (β, β') records, for every position in β, whether it holds an agent of this kind and, if so, whether the right position of that agent lies before β', inside β' (and at which offset), or after β'. The descriptor records the symmetric information for the positions in β'. It has $O(B \log B)$ bits. There are $O(n^2/B^2)$ descriptors, each computed in $O(B)$ time using the inverse permutations, so all of them take $O(n^2/B)$ time.

Descriptors keep the two active masks of a group consistent. When a member of G_k is removed through one side, we mark its bit in the chosen mask of that side and clear its active bit. If its position on the other side lies inside the other current bucket, the descriptor gives its offset there, and we clear that bit as well. If that position lies in a later bucket, nothing needs to be recorded, for the following reason. When the active mask of one side becomes zero, the head on that side lies in a later bucket, and we *refresh* that side: we move to the following buckets, one at a time, until we reach a bucket that contains a present member, and we compute its active mask. Every member removed through this side was the head at the time, so its position lies in an earlier bucket. Hence a member with a position in a newly entered bucket is present if and only if it has not been removed through the other side, that is, if and only if its position on the other side lies after the other current bucket, or inside it with its bit set in the other active mask. With a lookup table indexed by a descriptor and an active mask, each bucket takes $O(1)$ time. If no later bucket contains a present member, the group is empty. We flush both chosen masks, set both frontiers to n, and propagate its empty flag to both neighbours. Any pending refresh on its other side then needs no further scan. If both active masks become zero together, refresh the sides successively. The same rule remains valid when the other active mask is zero, since all present members then lie beyond its current bucket. Refreshing a bucket resets its chosen mask to zero. Bits already

flushed into its chosen vector are never changed.

Windows and the table. We cut $G_1, \dots, G_K$ into *windows* of $s \geq 3$ consecutive groups, separated by single *boundary groups*. A short final window is padded with empty groups, and a missing boundary is treated as empty. The local states of the groups of a window take $O(sB \log B)$ bits, so they fit into one machine word if $sB \log B \leq \varepsilon \log n$ for a suitable constant $\varepsilon > 0$. A universal table Φ is indexed by such a word together with the offsets of the two boundary heads facing the window, namely the right head of the boundary group before it and the left head of the boundary group after it. Empty groups have no heads and are never ready. The entry of Φ is the outcome of the following simulation: as long as every nonempty group of the window has active positions on both sides and some pair of adjacent groups inside the window is ready, match the heads of the leftmost ready pair and update the masks as described above. The entry consists of the final word and the list of (at most four) groups and sides whose active mask became zero. These need a refresh. The boundary groups do not change during the simulation, so every pair it matches is a pair of mutual first choices of the current market. Every simulated match clears active bits, so there are at most sB matches in a table entry. The table has $2^{O(sB \log B)}$ entries, each computed in time polynomial in sB by this fixed deterministic simulation. Invalid encodings can be assigned arbitrary outputs. The bucket-refresh table has $2^{O(B \log B)}$ entries and is smaller. Both tables are constructed as part of the algorithm.

Main loop. We keep a queue of windows and boundary groups, initially all of them in left-to-right order, suppressing duplicate queue entries with a flag. The invariant is that no window outside the queue contains a ready pair and no boundary group outside the queue is incident to a ready pair.

- To process a window, look up Φ and store the new word. Carry out every requested refresh: flush the chosen mask into the chosen vector, refresh the side as described above, load the new descriptor and masks, and update the bucket comparison and the flags of the neighbouring group on that side. If there was a refresh, put the window back into the queue. In any case, put the two adjacent boundary groups into the queue.
- To process a boundary group G_b, test the pairs G_{b-1}, G_b and G_b, G_{b+1} for readiness. If one of them is ready, match its heads, which changes G_b and one group in an adjacent window word with $O(1)$ word operations, refresh all current buckets that became empty, update the affected neighbour comparisons and empty flags, and put G_b and both adjacent windows into the queue.

Whether G_k and G_{k+1} are ready depends only on the frontiers of $G_{k-1}, \dots, G_{k+2}$. Hence a change in a group affects only pairs at distance at most two from it, and since $s \geq 3$, the steps above put every window and boundary group that contains such a pair into the queue. This maintains the invariant. When the queue is empty, no pair is ready, so by Lemma 2.1 no acceptable pair is left and the elimination is complete. Initialization loads the first buckets, refreshes any zero active masks, and computes the neighbour comparisons and flags. Each field update changes only $O(1)$ packed group records and therefore costs $O(1)$ word operations, apart from the bucket advances charged below. Chosen masks are stored into their global bit vectors using $O(1)$ word operations per flush, since B is smaller than a word. Finally, we flush all remaining chosen masks and read off sheet $2n + 1$.

5.3 Running time

Each frontier passes through at most $\lceil n/B \rceil + 1$ buckets, so all refreshes, including skipped buckets and initialization, take $O(K(n/B+1))$ time. A boundary group loses each of its at most n members once. There are $O(K/s)$ such groups, so there are $O(Kn/s)$ boundary matchings, each costing $O(1)$ time apart from refreshes.

A call of Φ ends either with a refresh or with no internal ready pair. After a completed call, the window is queued again only after a refresh or a matching at an adjacent boundary. Hence there are $O(K/s + Kn/B + Kn/s)$ table calls and the same bound on boundary processings. Including inverse permutations, descriptors, chosen vectors, and both tables, the total time is

$$O\left(\frac{n^2}{B} + \frac{Kn}{B} + \frac{Kn}{s} + K + 2^{O(sB\log B)}\,\mathrm{poly}(sB)\right). \tag{5.2}$$

Here the $O(Kn/\log n + K)$ words used by the chosen vectors are absorbed by the displayed terms. Each descriptor fits in a word under the same parameter choice as the window states.

Choose

$$B = s = \left\lfloor c\sqrt{\frac{\log n}{\log\log n}} \right\rfloor$$

for a sufficiently small positive constant c. Then $sB\log B = O(c^2\log n)$: the packed states fit in a word, and the tabulation term in (5.2) is $n^{\delta}\,\mathrm{polylog}\,n$ for some fixed $\delta < 1$. Since $K = 4R = 16n+8$, the total time is

$$O\left(n^2\sqrt{\frac{\log\log n}{\log n}}\right).$$

Together with Theorem 4.11 and the $O(n)$-time extraction of the middle sheet, this proves Theorem 1.1.

6 Concluding remarks

Two-list stable matching admits a deterministic $o(n^2)$ algorithm by unrolling the cycle of groups into a path of $O(n)$ sheets. The principal question is whether a strongly subquadratic, or even near-linear, bound is possible. Our implementation batches the elimination of a market with $\Theta(n^2)$ agents. A stronger speedup within this approach may require a more compact representation of how a block of sheets transforms the frontiers, rather than simulating all of its eliminations.

The linear-time stability verifier (Künnemann et al., 2019) remains an obstacle to standard deterministic SETH-based lower bounds: the non-reducibility framework of Carmosino et al. (2016) would turn a superlinear lower bound of this form into a refutation of NSETH. Other hypotheses or types of reductions may be more suitable.

For $d \geq 3$ lists, the group graph is $K_{d,d}$ rather than a cycle. It can be unrolled into a tree, but the number of copies at depth $\Theta(n)$ grows exponentially. Extending the end-bound argument without this growth may provide a route to an $o(n^2)$ algorithm for larger fixed d.

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A The seam-minimizing stable matching

We identify the matching μ returned by the algorithm. For every depth R, define the *count matrix* of ν_R by

$$c_R(m,w) = \big|\{a \in [R] : \nu_R(m^a) = w\}\big|.$$

Monotonicity makes these counts determine the matching. A linear description of the counts gives an exact certificate for the seam objective.

A.1 Counts and stability

For a nonnegative integer matrix c indexed by $M \times W$, set

$$A_c(m,w) = \sum_{w' \succeq_m w} c(m,w'), \qquad B_c(m,w) = \sum_{m' \succ_w m} c(m',w).$$

At depth R, define the *slack*

$$s_c^{(R)}(m,w) = A_c(m,w) + B_c(m,w) + \kappa(m,w) - R. \tag{A.1}$$

A pair is *tight* if its slack is zero. For $c = c_R$, the quantity $A_c(m,w)$ counts the copies of m with a partner at least as good as w, while $B_c(m,w)$ counts the copies of w with a partner strictly better than m.

Lemma A.1 (Counts)**.** *The count matrix c_R is the unique nonnegative integer matrix c whose row and column sums are at most R and that satisfies*

$$s_c^{(R)}(m,w) \geq 0 \quad \textit{for every } (m,w), \qquad c(m,w) > 0 \implies s_c^{(R)}(m,w) = 0.$$

Proof. Let ν be a monotone matching of $\mathcal{C}_R$, and let c be its count matrix. Fix (m,w), and abbreviate $A = A_c(m,w)$, $B = B_c(m,w)$, and $k = \kappa(m,w)$. The copies of m whose partner is worse than w or unmatched occupy exactly the sheets $a < R - A$. The copies of w whose partner is strictly better than m occupy exactly the sheets $b < B$.

An acceptable pair (m^a, w^{a-k}), with $k \leq a < R$, blocks precisely when $a < R - A$ and $a - k \geq B$. Indeed, in that case the woman cannot be matched to a copy of m: the only such acceptable copy is m^a, whose partner is worse than w. A blocking copy of (m,w) therefore exists exactly when $B + k < R - A$, or $s_c^{(R)}(m,w) < 0$. Thus a monotone matching is stable if and only if all its slacks are nonnegative.

If $c(m,w) > 0$, the man's copies matched to w occupy

$$[R - A,\ R - A + c(m,w)),$$

whereas the woman's copies matched to m occupy $[B,\ B + c(m,w))$. The matching edges shift these intervals by k, so $R - A = B + k$ and the pair is tight. Applying this to ν_R, which is monotone by Lemma 3.3, proves necessity.

Conversely, suppose c satisfies the stated conditions. Assign partner w to the copies m^a with

$$a \in [R - A_c(m,w),\ R - A_c(m,w) + c(m,w)),$$

and partner m to the copies w^b with

$$b \in [B_c(m,w),\ B_c(m,w) + c(m,w)).$$

For a fixed man the intervals are disjoint and fill $[R - \sum_w c(m,w), R)$ from worst to best partner. For a fixed woman they are disjoint and fill $[0, \sum_m c(m,w))$ from best to worst partner. Leave the remaining copies unmatched. Tightness on every positive entry gives

$$R - A_c(m,w) = B_c(m,w) + \kappa(m,w),$$

so the two assignments agree and use acceptable pairs. They define a monotone matching with count matrix c. Its nonnegative slacks imply stability, and uniqueness in Corollary 3.2 identifies it with ν_R. □

A.2 An exact gap identity

Let

$$D_R = nR - \sum_{m,w} c_R(m,w)$$

be the number of unmatched men in ν_R. For a stable matching η of the original instance, define

$$b_\eta(m,w) = [\eta(m) \succeq_m w] + [\eta^{-1}(w) \succ_w m] - 1, \tag{A.2}$$

where brackets denote indicators. Stability of η gives $b_\eta(m,w) \geq 0$ for every pair. It is zero on an edge of η, and on a non-edge it is one precisely when both endpoints prefer their η-partners to each other.

Lemma A.2 (Gap identity). *For every depth R and every stable matching η of the original instance,*

$$q(\eta) - D_R = \sum_{m,w} c_R(m,w) b_\eta(m,w) + \sum_m s_{c_R}^{(R)}(m, \eta(m)). \tag{A.3}$$

Both sums on the right are nonnegative.

Proof. Put $c = c_R$. When $\sum_m (A_c(m,\eta(m)) + B_c(m,\eta(m)))$ is expanded, the coefficient of $c(m,w)$ is

$$[w \succeq_m \eta(m)] + [m \succ_w \eta^{-1}(w)] = 1 - b_\eta(m,w).$$

This identity follows from strict preferences. It also holds when $w = \eta(m)$, when both sides equal one. Consequently,

$$\begin{aligned} \sum_m s_c^{(R)}(m,\eta(m)) &= \sum_{m,w} c(m,w)(1 - b_\eta(m,w)) + q(\eta) - nR \\ &= q(\eta) - D_R - \sum_{m,w} c(m,w) b_\eta(m,w). \end{aligned}$$

Rearranging gives (A.3). Nonnegativity follows from stability of η and Lemma A.1. □

Proposition A.3. *The stationary matching μ, and hence the matching returned by the algorithm, is the unique stable matching minimizing $q(\eta) = |\eta \cap (M_0 \times W_1)|$.*

Proof. Take the fixed long cover of Section 3.2, and write $c = c_L$. By Lemma 4.1, $D_L = q(\mu)$. The nonnegative right-hand side of (A.3) therefore gives $q(\eta) \geq q(\mu)$ for every stable η.

Suppose equality holds. Then both sums in (A.3) vanish. Hence every edge of η is tight for c at depth L, and $b_\eta(m, w) = 0$ wherever $c(m, w) > 0$. The same statements hold for μ.

For $\rho \in \{\eta, \mu\}$, let $\mathbf{1}_\rho$ be its incidence matrix. The slack of $c + \mathbf{1}_\rho$ at depth $L + 1$ is

$$s^{(L+1)}_{c+\mathbf{1}_\rho}(m, w) = s^{(L)}_c(m, w) + b_\rho(m, w). \tag{A.4}$$

Every slack is nonnegative. A positive entry inherited from c remains tight, since both terms on the right are zero there. Every edge of ρ is also tight, because it is tight for c and b_ρ is zero on that edge. Row and column sums are at most $L + 1$. By Lemma A.1,

$$c + \mathbf{1}_\eta = c_{L+1} = c + \mathbf{1}_\mu.$$

Thus $\eta = \mu$. □

Corollary A.4. *For any $i, j \in \{0, 1\}$, the unique stable matching minimizing or maximizing $|\eta \cap (M_i \times W_j)|$ can be found within the bound of Theorem 1.1.*

Proof. Relabel the types to send the desired block to $M_0 \times W_1$ and apply the algorithm. For maximization, use

$$|\eta \cap (M_i \times W_j)| = |M_i| - |\eta \cap (M_i \times W_{1-j})|$$

and minimize the complementary block. Uniqueness follows from Proposition A.3 in the relabelled instance. □